Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
kilicgokturk@kilicgokturk:~$ ls /etc/ > file.txt
kilicgokturk@kilicgokturk:~$ ls >> file.txt
kilicgokturk@kilicgokturk:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
```

Рисунок 1: Запись в файл

Работа фильтра

```
kilicgokturk@kilicgokturk:~$
kilicgokturk@kilicgokturk:~$ grep .conf file.txt > conf.txt
kilicgokturk@kilicgokturk:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/kilicgokturk/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a874
86645945
/home/kilicgokturk/.npm/_cacache/index-v5/ce
/home/kilicgokturk/.npm/_cacache/index-v5/09/c0
/home/kilicgokturk/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def302
20242c53
/home/kilicgokturk/.npm/_cacache/index-v5/84/ce
/home/kilicgokturk/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f
59e2edcc
/home/kilicgokturk/.npm/_cacache/index-v5/cd
/home/kilicgokturk/.npm/_cacache/index-v5/c9
/home/kilicgokturk/.npm/_cacache/index-v5/c0
/home/kilicgokturk/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d9
90504757
/home/kilicgokturk/.npm/_cacache/index-v5/9f/ae/c0778548017b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b20
3217f693
/home/kilicgokturk/.npm/_cacache/index-v5/af/cf
/home/kilicgokturk/.npm/_cacache/index-v5/c3
/home/kilicgokturk/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db8
861160d3
/home/kilicgokturk/.npm/_cacache/index-v5/ca
/home/kilicgokturk/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554
f3f3ba2e
/home/kilicgokturk/snap/hugo/common
/home/kilicgokturk/snap/hugo/current
/home/kilicgokturk/conf.txt
kilicgokturk@kilicgokturk:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
kilicgokturk@kilicgokturk:~$
kilicgokturk@kilicgokturk:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
kilicgokturk@kilicgokturk:~$
kilicgokturk@kilicgokturk:~$ find ~ -name "log*" > logfile &
[1] 6716
kilicgokturk@kilicgokturk:~$
[1]+  Done                    find ~ -name "log*" > logfile
kilicgokturk@kilicgokturk:~$ rm logfile
kilicgokturk@kilicgokturk:~$
kilicgokturk@kilicgokturk:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
kilicgokturk@kilicgokturk:~$
kilicgokturk@kilicgokturk:~$ gedit &
[1] 6762
kilicgokturk@kilicgokturk:~$ ps | grep gedit
   6762 pts/1    00:00:00 gedit
kilicgokturk@kilicgokturk:~$ kill 6762
[1]+  Terminated              gedit
kilicgokturk@kilicgokturk:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of df.  df displays the amount of space available
       on  the file system containing each file name argument.  If no file name is given, the space
       available on all currently mounted file systems is shown.  Space is shown in  1K  blocks  by
       default,  unless  the  environment  variable  POSIXLY_CORRECT is set, in which case 512-byte
       blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file  system,
       df  shows  the space available on that file system rather than on the file system containing
       the device node.  This version of df cannot show the space available on unmounted file  sys-
       tems,  because on most kinds of systems doing so requires non-portable intimate knowledge of
       file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all  file  systems  by
       default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes  by  SIZE  before  printing  them;  e.g., '-BM' prints sizes in units of
              1,048,576 bytes; see SIZE format below
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent sizes rather than device usage; although the apparent size is usually
              smaller, it may be larger due to holes in ('sparse') files,  internal  fragmentation,
              indirect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by  SIZE  before  printing  them;  e.g., '-BM' prints sizes in units of
              1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
kilicgokturk@kilicgokturk:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 90122600  64410968  59% /
devtmpfs         4011280        0   4011280   0% /dev
tmpfs            4035656      100   4035556   1% /dev/shm
tmpfs            1614264     1968   1612296   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035656     1644   4034012   1% /tmp
/dev/nvme0n1p3 155186176 90122600  64410968  59% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807128      176    806952   1% /run/user/1040
/dev/sr0         2677920  2677920         0 100% /run/media/kilicgokturk/Fedora-WS-Live-43
kilicgokturk@kilicgokturk:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4         ./.npm/_cacache/index-v5/bf
4         ./.npm/_cacache/index-v5/30/1a
4         ./.npm/_cacache/index-v5/30
424       ./.npm/_cacache/index-v5
115600    ./.npm/_cacache
115664    ./.npm
16        ./snap/hugo/25676/.config/go/telemetry/local
0         ./snap/hugo/25676/.config/go/telemetry/upload
16        ./snap/hugo/25676/.config/go/telemetry
16        ./snap/hugo/25676/.config/go
16        ./snap/hugo/25676/.config
16        ./snap/hugo/25676
0         ./snap/hugo/common
20        ./snap/hugo
20        ./snap
0         ./monthly
0         ./reports/monthly/monthly
0         ./reports/monthly
0         ./reports
4         ./ski.plases/equipment
0         ./ski.plases/plans
4         ./ski.plases
0         ./australia
0         ./play/games/play
0         ./play/games
0         ./play
1491316   .
kilicgokturk@kilicgokturk:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/kilicgokturk/.npm/_cacache/index-v5/5274d
/home/kilicgokturk/.npm/_cacache/index-v5/2b
/home/kilicgokturk/.npm/_cacache/index-v5/2b/8e
/home/kilicgokturk/.npm/_cacache/index-v5/69
/home/kilicgokturk/.npm/_cacache/index-v5/69/e3
/home/kilicgokturk/.npm/_cacache/index-v5/9c
/home/kilicgokturk/.npm/_cacache/index-v5/9c/e7
/home/kilicgokturk/.npm/_cacache/index-v5/bf
/home/kilicgokturk/.npm/_cacache/index-v5/bf/07
/home/kilicgokturk/.npm/_cacache/index-v5/30
/home/kilicgokturk/.npm/_cacache/index-v5/30/1a
/home/kilicgokturk/snap
/home/kilicgokturk/snap/hugo
/home/kilicgokturk/snap/hugo/25676
/home/kilicgokturk/snap/hugo/25676/.config
/home/kilicgokturk/snap/hugo/25676/.config/go
/home/kilicgokturk/snap/hugo/25676/.config/go/telemetry
/home/kilicgokturk/snap/hugo/25676/.config/go/telemetry/local
/home/kilicgokturk/snap/hugo/25676/.config/go/telemetry/upload
/home/kilicgokturk/snap/hugo/common
/home/kilicgokturk/monthly
/home/kilicgokturk/reports
/home/kilicgokturk/reports/monthly
/home/kilicgokturk/reports/monthly/monthly
/home/kilicgokturk/ski.plases
/home/kilicgokturk/ski.plases/equipment
/home/kilicgokturk/ski.plases/plans
/home/kilicgokturk/australia
/home/kilicgokturk/play
/home/kilicgokturk/play/games
/home/kilicgokturk/play/games/play
kilicgokturk@kilicgokturk:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.